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Question 1

When transmitting a document electronically, one may either scan and e-mail the document image or apply **optical character recognition** (OCR) and send the resulting text file. While scanning preserves the complete visual information of the document, OCR attempts to interpret the image content and convert it into machine-readable text. As discussed in the **Handbook of Document Image Processing and Recognition**, OCR accuracy depends strongly on document image quality, and errors introduced during binarization and page segmentation can significantly degrade recognition performance. Therefore, the choice between these two approaches depends on the expected **OCR error rate, computational cost, and the quality of the document image**.

Scan and E-mail the Image

Advantages

- Preserves the complete visual information of the document.
- Avoids recognition errors caused by OCR, binarization, or segmentation.
- Suitable for low-quality documents with noise, broken, or touching characters.

Disadvantages

- Larger file size compared to text.
- Content is not directly searchable or editable.
- Manual effort is required for text extraction.

OCR and Send the Text File

Advantages

- Produces searchable and editable text.
- Smaller file size and efficient storage.

- Enables automated text processing and analysis.

Disadvantages

- Highly sensitive to document image quality.
- Errors due to binarization and page segmentation can propagate to OCR output.
- Low-quality documents may require costly post-processing or manual correction [1].

When Is Each Approach Preferable?

OCR is preferable when the document is **cleanly printed, has good contrast**, and is expected to produce a low OCR error rate, making automatic recognition and correction cost-effective. Scanning and e-mailing the image is preferable when the **document quality is poor**, when OCR error rates are likely to be high, or when preserving the exact visual appearance of the document is important [1].

Question 2

(a) What is in a junk e-mail that lets us know it is junk?

Junk e-mails (spam) typically contain recurring lexical, stylistic, and structural patterns that distinguish them from legitimate messages. Common indicators include promotional or deceptive keywords such as *free*, *urgent*, or *limited offer*, excessive capitalization or unusual punctuation, and spelling obfuscations designed to evade simple keyword filters. Spam messages often contain a large number of hyperlinks, unsolicited attachments, or HTML-heavy formatting, and they frequently lack meaningful personalization. Although individual features may also appear in legitimate e-mails, the combination and frequency of these characteristics provide strong statistical evidence that an e-mail is junk.

(b) How can the computer detect junk through a syntactic analysis?

Through syntactic analysis, the computer detects junk e-mail by examining the form and structure of the text rather than its semantic meaning. Incoming e-mails are first preprocessed by removing headers, HTML tags, punctuation, and by normalizing the text. Each message is then represented as a feature vector using syntactic features such as word presence, word frequency, TF-IDF weights, or n-grams. Supervised learning algorithms, such as Naïve Bayes or Support Vector Machines, learn statistical regularities that differentiate spam from legitimate e-mails. For a new message, the classifier evaluates its syntactic feature pattern and assigns it to the spam or non-spam class based on the learned model.

(c) What should the computer do if it detects a junk e-mail?

The appropriate action after detecting junk e-mail depends on the cost of misclassification. Incorrectly blocking a legitimate message is generally more costly than allowing a spam message to pass. Automatically deleting detected spam is therefore risky, as it may result in the permanent loss of important e-mails. A more practical approach is to move suspected junk e-mails to a separate folder, allowing users to review and recover misclassified messages. Alternatively, the system may highlight or flag suspected spam, leaving the final decision to the user. In practice, moving or flagging junk e-mail is preferred over automatic deletion due to the high cost of false positives.

Question 3

(a) Define the constraints

An automated taxi operates under multiple constraints. Safety constraints require the vehicle to avoid collisions and ensure passenger and pedestrian safety in uncertain and dynamic environments. Physical constraints arise from the vehicle's mechanical limits such as speed, acceleration, braking capability, and sensor range. Legal and regulatory constraints include traffic laws, road rules, and jurisdiction-specific autonomous vehicle regulations. Environmental constraints involve road conditions, traffic congestion, weather, and incomplete or noisy sensor data. In addition, operational constraints such as passenger wait times, routing efficiency, and fleet-level balancing must be satisfied, particularly in robo-taxi systems.

(b) What are the inputs?

The automated taxi relies on continuous streams of input data, including perception inputs from cameras, LiDAR, radar, GPS, and inertial sensors to understand its surroundings. Navigation inputs include digital maps, lane-level information, traffic signals, and real-time traffic conditions. Passenger inputs include destination requests, ride confirmations, preferences, and optional voice commands. In fleet-based settings, system-level inputs such as demand distribution, vehicle availability, and shared traffic information are also used to improve coordination and efficiency.

(c) What is the output?

The output of an automated taxi consists of real-time decisions and control actions. These include low-level vehicle control commands such as steering, acceleration, braking, and signaling, as well as higher-level navigation decisions such as route selection, lane changes, stops, and drop-off actions. User-facing outputs may include estimated arrival times, route updates, and safety or status notifications provided to the passenger.

(d) How can we communicate with the passenger?

Communication with passengers must be clear, intuitive, and reliable. This can be achieved through visual interfaces such as touchscreens or mobile applications for destination entry and trip monitoring. Audio and speech-based interfaces allow natural interaction through spoken commands and system announcements, enabling hands-free communication. Secure authentication mechanisms, such as app-based login or biometric verification, ensure that the correct passenger is authorized to access the vehicle. Effective communication increases user trust and usability in autonomous taxi systems.

(e) Do we need to communicate with other automated taxis? Do we need a “language”?

Communication between automated taxis is highly beneficial, especially in fleet-based systems. Sharing information such as traffic conditions, road hazards, vehicle availability, and demand patterns enables coordinated routing and efficient fleet management. However, this communication does not require a human-like natural language. Instead, it relies on structured data exchange, standardized protocols, and shared representations. Such machine-level communication allows cooperation without ambiguity and improves overall system performance.

Question 4

(a) What are the parameters?

A circle in a two-dimensional plane is completely defined by its center and radius. The parameters of a circle hypothesis are:

- The center coordinates (x_c, y_c)
- The radius r

Thus, the hypothesis class of circles has three parameters:

$$\theta = (x_c, y_c, r)$$

and can be represented by the implicit equation

$$(x - x_c)^2 + (y - y_c)^2 = r^2.$$

(b) How can the parameters of a circle hypothesis be calculated?

The parameters of a circle hypothesis can be estimated from data points using curve fitting techniques. In the noise-free case, three non-collinear points uniquely determine a circle. In practical scenarios with noisy data, the parameters are typically estimated by minimizing an error function. Common approaches include:

- Algebraic least-squares fitting by expanding the circle equation and solving a linear system.
- Geometric fitting by minimizing the sum of squared distances from the data points to the circle.

These methods allow the circle parameters to be estimated even when the data does not lie exactly on a perfect circle.

(c) What if the hypothesis class is an ellipse?

An ellipse is a more general geometric shape than a circle and therefore has more parameters. An ellipse is defined by:

- Center coordinates (x_c, y_c)
- Semi-major axis length a
- Semi-minor axis length b
- Orientation angle θ

Hence, an ellipse hypothesis has five parameters. A general implicit form of an ellipse is given by

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

with appropriate constraints to ensure the conic section is an ellipse. Estimating these parameters typically involves constrained optimization or least-squares fitting methods.

(d) Why does it make more sense to use an ellipse instead of a circle?

Using an ellipse instead of a circle often makes more sense because it provides a more expressive hypothesis class. A circle assumes that the data has the same spread in all directions, whereas an ellipse allows different spreads along different axes. This added flexibility enables ellipses to better model elongated or anisotropic data distributions. From a learning perspective, restricting the hypothesis class to circles may introduce high bias, while ellipses offer a better balance between model simplicity and representational power.

Question 5

A manufacturer states that the Z-Phone smartphone has a mean consumer life of $\mu = 42$ months with a standard deviation of $\sigma = 8$ months. Assuming a normal distribution, we compute the probability that a randomly selected Z-Phone lasts between 20 and 30 months.

Solution

Let X denote the lifetime (in months) of a Z-Phone. Then,

$$X \sim \mathcal{N}(42, 8^2).$$

We are required to compute:

$$P(20 \leq X \leq 30).$$

First, we standardize the random variable using the transformation

$$Z = \frac{X - \mu}{\sigma}.$$

For $X = 20$,

$$z_1 = \frac{20 - 42}{8} = -2.75,$$

and for $X = 30$,

$$z_2 = \frac{30 - 42}{8} = -1.5.$$

Thus, the required probability becomes

$$P(20 \leq X \leq 30) = P(-2.75 \leq Z \leq -1.5).$$

Using the cumulative distribution function of the standard normal distribution,

$$P(-2.75 \leq Z \leq -1.5) = \Phi(-1.5) - \Phi(-2.75),$$

where

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

The numerical values of $\Phi(-1.5)$ and $\Phi(-2.75)$ were computed using Python in the submitted Jupyter notebook:

$$\Phi(-1.5) \approx 0.0668, \quad \Phi(-2.75) \approx 0.0030.$$

Therefore,

$$P(20 \leq X \leq 30) \approx 0.0668 - 0.0030 = 0.0638.$$

Final Answer

$$\boxed{P(20 \leq X \leq 30) \approx 0.064}$$

Hence, there is approximately a 6.4% probability that a randomly selected Z-Phone will last between 20 and 30 months.

Question 6

An experiment was conducted to investigate the survival time (in hours) of an electronic component under accelerated life testing conditions. Eight components were tested, yielding the following observations:

$$75, 63, 100^+, 36, 51, 45, 80, 90,$$

where the observation 100^+ indicates that the unit was still functioning after 100 hours.

Nature of the Data

The observation 100^+ represents a *right-censored* survival time, meaning that the exact failure time of that component is unknown and is only known to exceed 100 hours. As discussed in standard survival analysis, right-censoring prevents the reliable computation of measures that depend on exact failure times, such as the arithmetic mean [10].

Meaningful Measure of Location

In the presence of right-censored data, a meaningful and well-defined measure of location is the **median survival time**. The median is defined as the time t at which the survivor function satisfies

$$S(t) = P(T > t) \leq 0.5.$$

This definition depends only on the ordering of failure times and the risk set, and does not require knowledge of exact failure times beyond the censoring point [10, 11].

Estimation Using the Kaplan–Meier Estimator

The Kaplan–Meier estimator is the standard nonparametric estimator of the survivor function in the presence of right-censored data [10, 11]. Ordering the observations gives:

$$36, 45, 51, 63, 75, 80, 90, 100^+.$$

At each observed failure time, the Kaplan–Meier estimator updates the estimated survival probability. After the failure at 63 hours, the estimated survivor function drops to 0.5. Therefore, the Kaplan–Meier estimate of the median survival time is the smallest time at which the estimated survivor function is less than or equal to 0.5.

Final Answer

Yes, a meaningful measure of location can be calculated. The appropriate measure is the *median survival time*, and its numerical value is

63 hours.

The mean survival time cannot be reliably estimated due to the presence of right-censored data.

Question 7

A gasoline manufacturer is investigating the cold start ignition time of an automobile engine.

(i) Descriptive Statistics and Box Plot for Formulation 1

The following ignition times (in seconds) were observed for a test vehicle:

1.75, 1.92, 2.62, 2.35, 3.09, 3.15, 2.53, 1.91.

Let the data be denoted by $x_1, x_2, \dots, x_n$ with $n = 8$.

The sample mean is given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample variance is

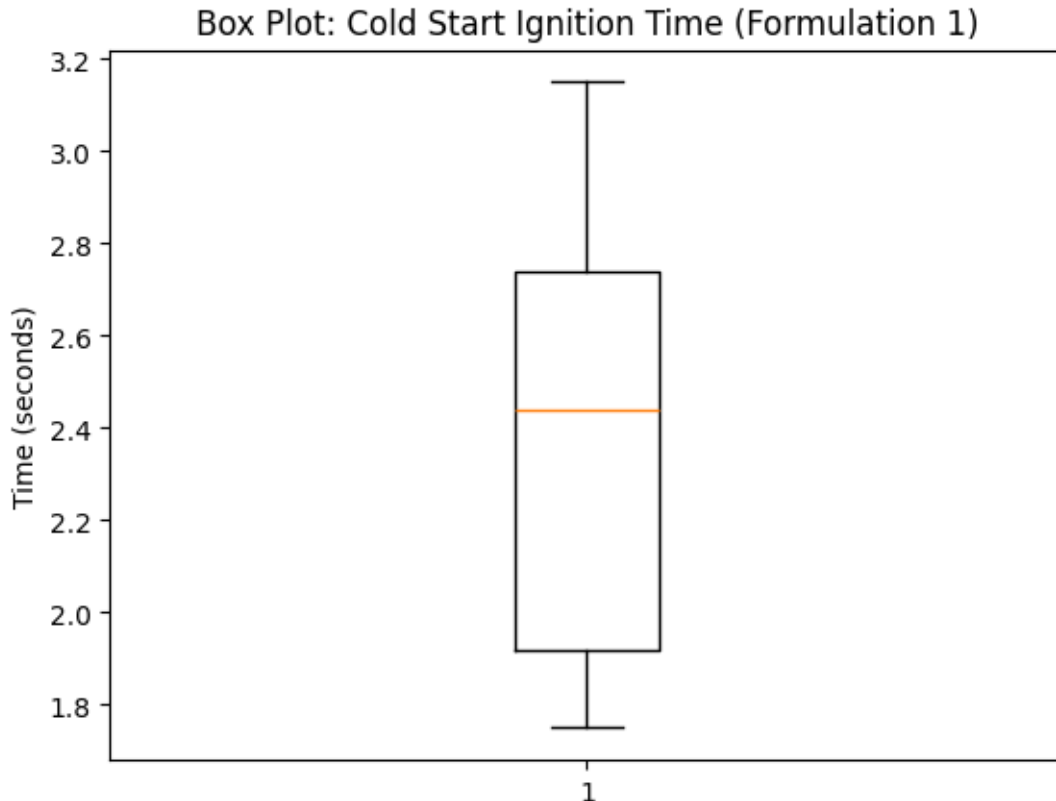
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2,$$

and the sample standard deviation is

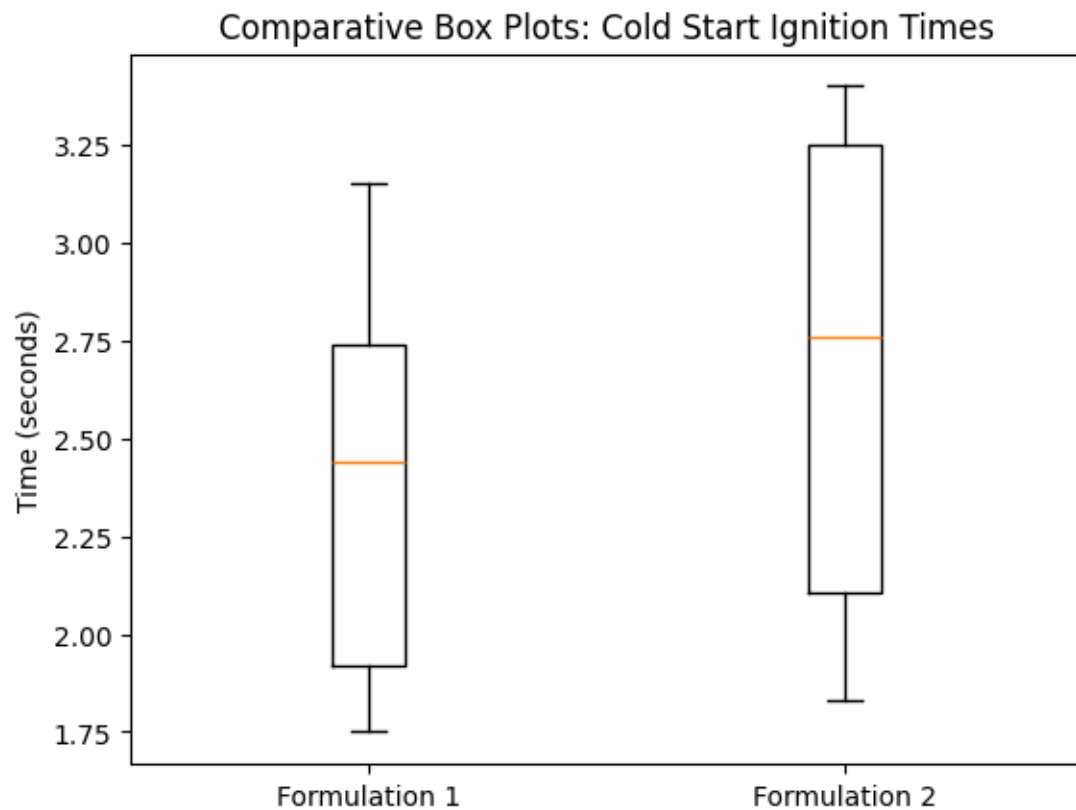
$$s = \sqrt{s^2}.$$

Using Python in a Jupyter Notebook, the following values were obtained:

$$\bar{x}_1 = 2.415 \text{ s}, \quad s_1^2 = 0.2854 \text{ s}^2, \quad s_1 = 0.5342 \text{ s}.$$



(ii) Comparative Box Plots for Two Gasoline Formulations



The box plots show that the second formulation has a higher median ignition time and greater variability than the first formulation. This indicates that the first formulation provides faster and more consistent cold start ignition performance.

Question 8

(a) Min–Max Normalization of Weight

Min–max normalization rescales a variable into the range $[0, 1]$ using the formula

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}},$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the variable.

From the data, the minimum and maximum weights are:

$$x_{\min} = 41 \text{ kg}, \quad x_{\max} = 136 \text{ kg}.$$

Applying min–max normalization to the weight variable yields a new column *Weight_norm*. For example, for P. Lee:

$$\frac{50 - 41}{136 - 41} = 0.094737.$$

(b) Binning the Weight Variable

The weight variable is discretized into three categories:

- **Low:** weight < 60 kg
- **Medium:** $60 \leq \text{weight} \leq 100$ kg
- **High:** weight > 100 kg

This produces a new categorical column *Weight_bin*, which can be useful for rule-based or interpretable models.

(c) Body Mass Index (BMI)

Body Mass Index (BMI) is an aggregated feature defined as

$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height (m)}^2}.$$

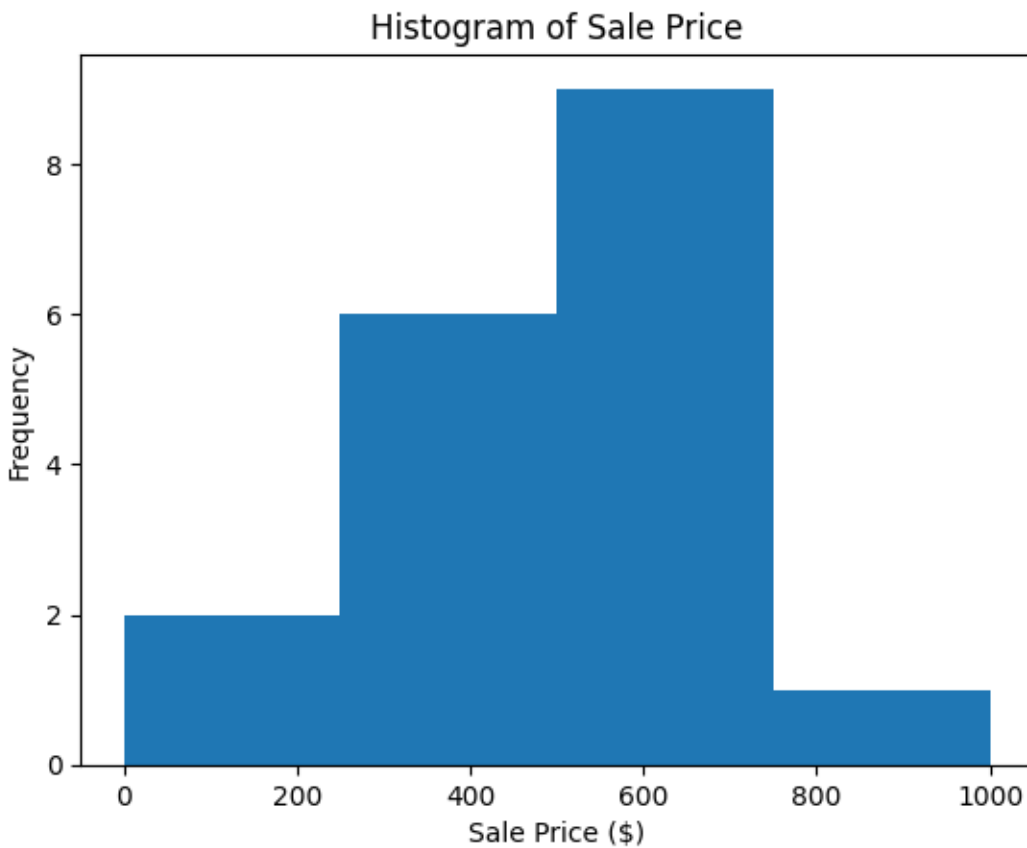
A new column *BMI* is computed for each patient using this formula.

Final Table with Engineered Features

	Name	Weight (kg)	Height (m)	Diabetes	Weight_norm	Weight_bin	BMI
0	P. Lee	50	1.52	0	0.094737	Low	21.641274
1	R. Jones	115	1.77	1	0.778947	High	36.707204
2	J. Smith	96	1.83	0	0.578947	Medium	28.666129
3	A. Patel	41	1.55	0	0.000000	Low	17.065557
4	M. Owen	79	1.82	0	0.400000	Medium	23.849777
5	S. Green	109	1.89	1	0.715789	High	30.514263
6	N. Cook	73	1.76	0	0.336842	Medium	23.566632
7	W. Hands	104	1.71	1	0.663158	High	35.566499
8	P. Rice	64	1.74	0	0.242105	Medium	21.138856
9	F. Marsh	136	1.78	1	1.000000	High	42.923873

Question 9

(a) Histogram



(b) Contingency table

Contingency Table (Store x Product Category):				
Product Category	Desktop	Laptop	Printer	Scanner
Store				
New York, NY	3	1	2	4
Washington, DC	2	2	2	2

(c) Summary tables by grouping

c-i) Grouped by Customer:

	Count	Total_Sales
Customer		
B. March	3	1700
E. Sims	1	700
G. Hinton	4	2150
H. Fu	1	450
H. Taylor	1	400
J. Bain	1	500
L. Nye	2	900
P. Judd	2	900
S. Cann	1	600
T. Goss	2	750

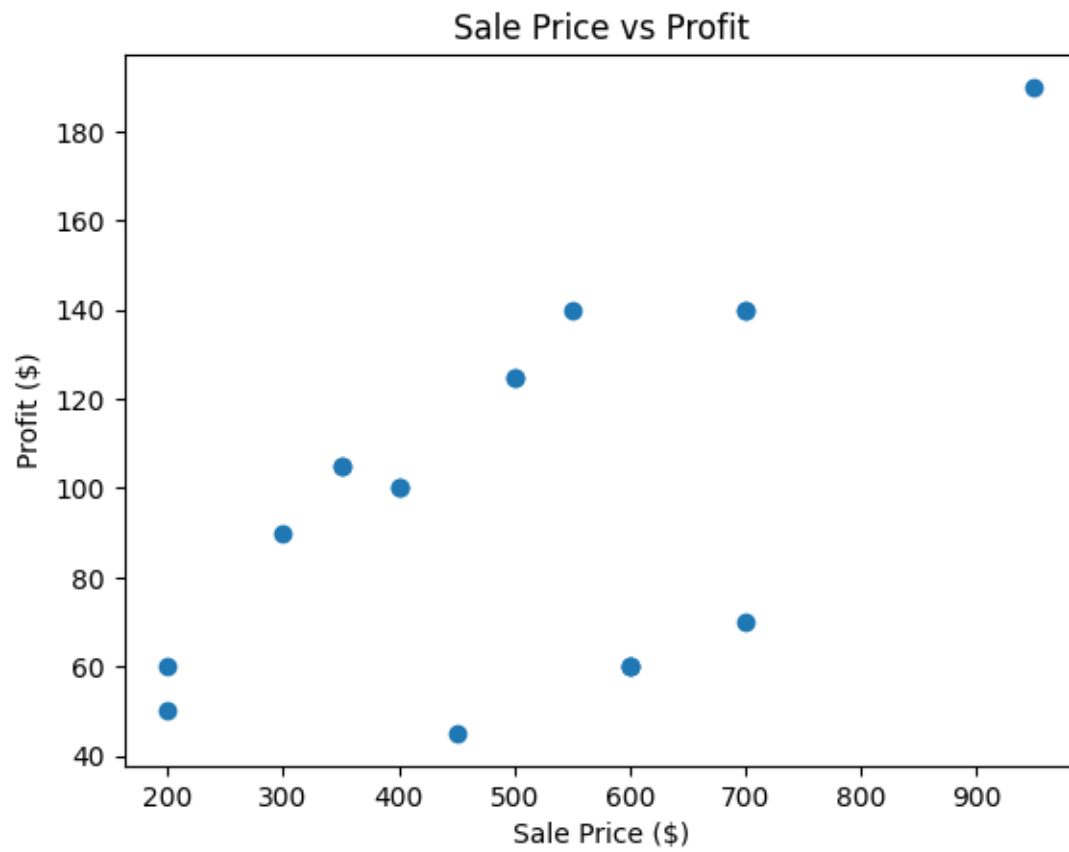
c-ii) Grouped by Store:

	Count	Mean_Sale_Price
Store		
New York, NY	10	485.0
Washington, DC	8	525.0

c-iii) Grouped by Product Category:

	Count	Total_Profit
Product Category		
Desktop	5	295
Laptop	3	470
Printer	4	360
Scanner	6	640

(d) Scatterplot



Question 10

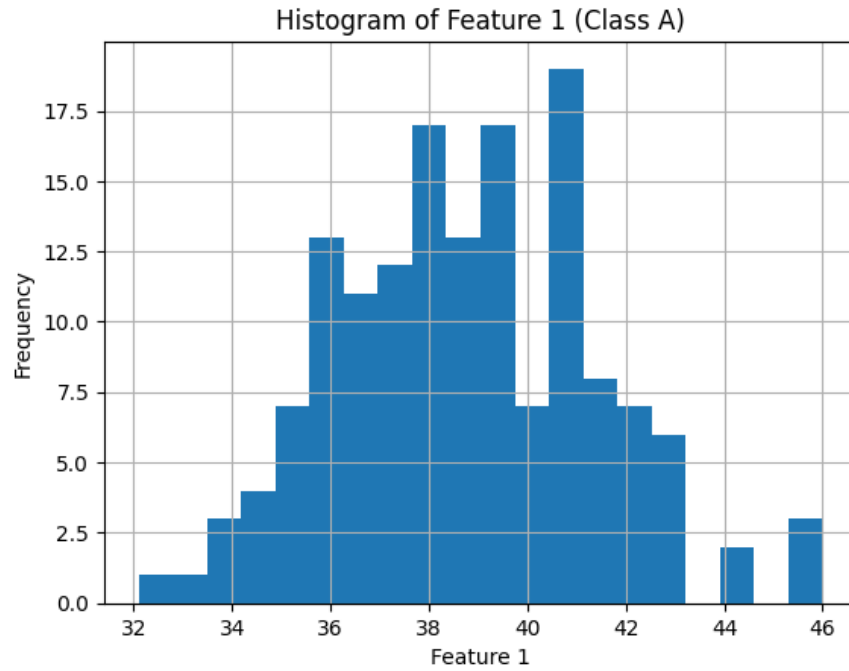
(a) Frequency of samples for each class

- Class A : 151
- Class B : 123
- Class C : 68

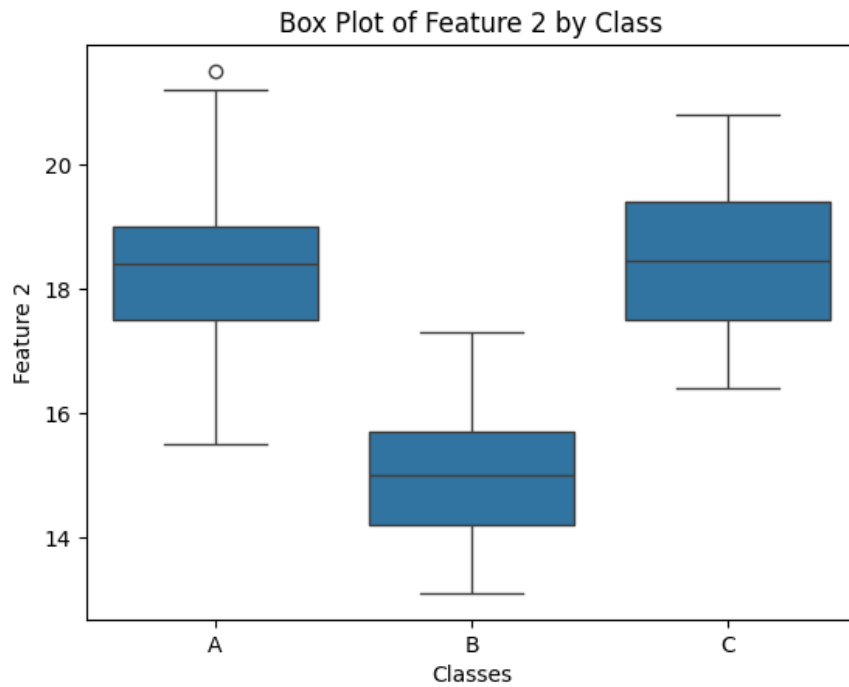
(b) Data description and IQR

Data Description:				
	Feature 1	Feature 2	Feature 3	Feature 4
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000
Interquartile Range (IQR):				
Feature 1	9.275			
Feature 2	3.100			
Feature 3	23.000			
Feature 4	1200.000			

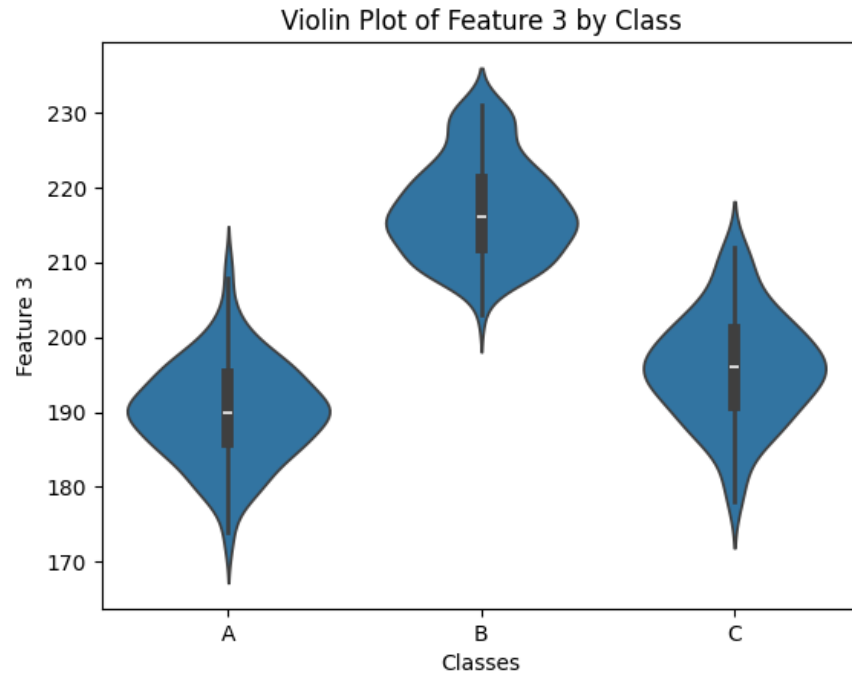
(c) Histogram of feature 1 for class A



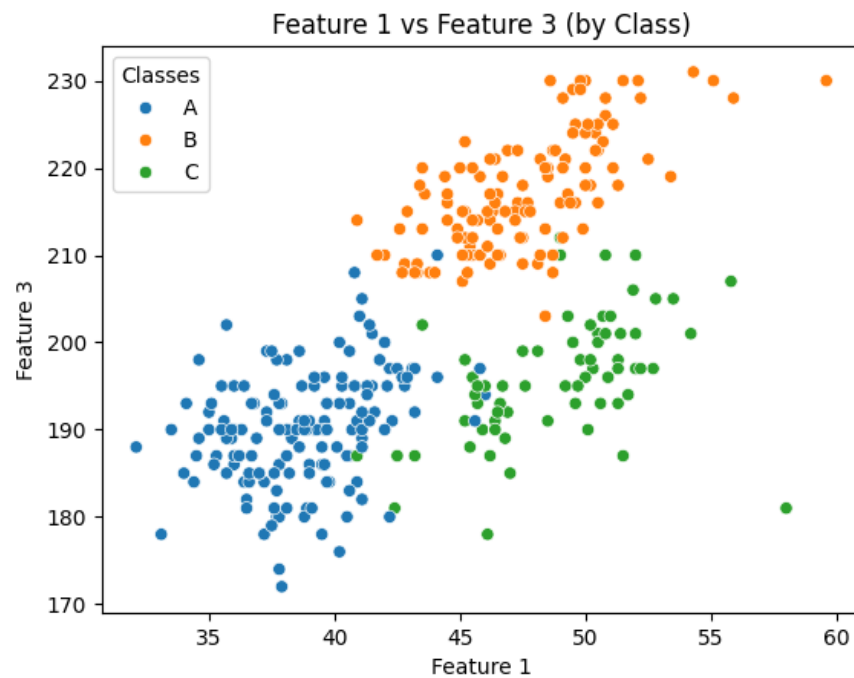
(d) Box plot of Feature 2 for each class



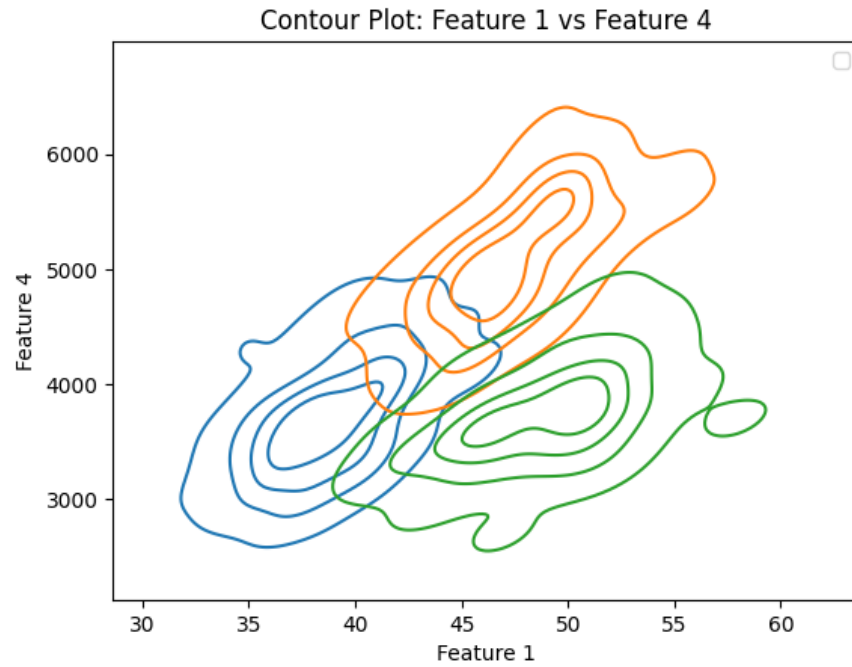
(e) Violin plot of Feature 3 for each class



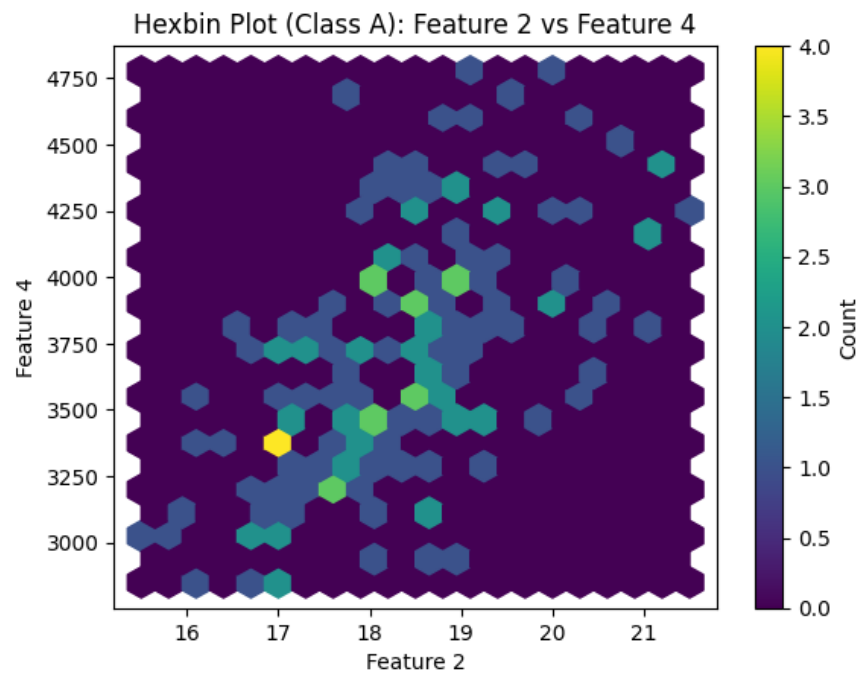
(f) Scatter plot: Feature 1 vs Feature 3



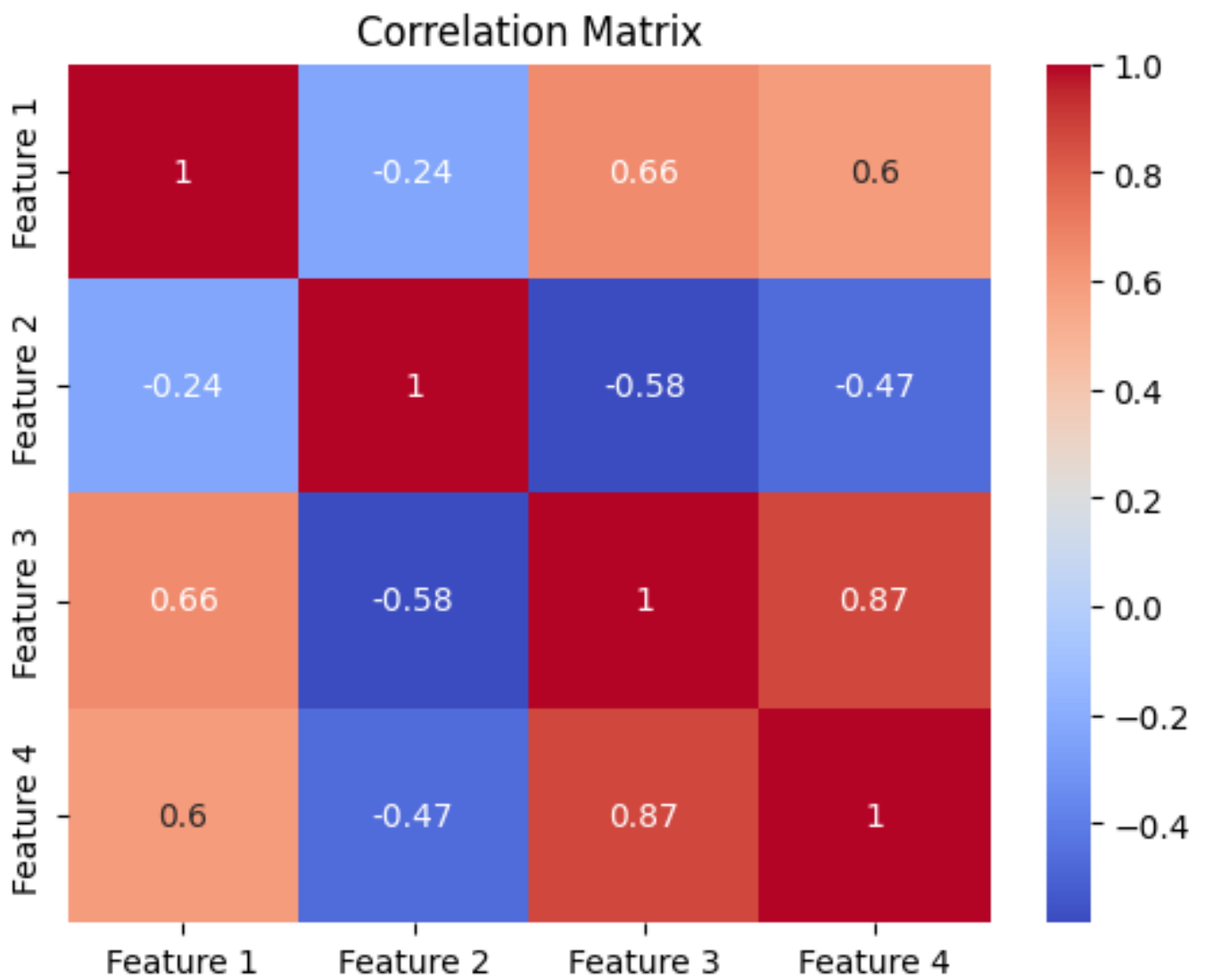
(g) Contour plot:



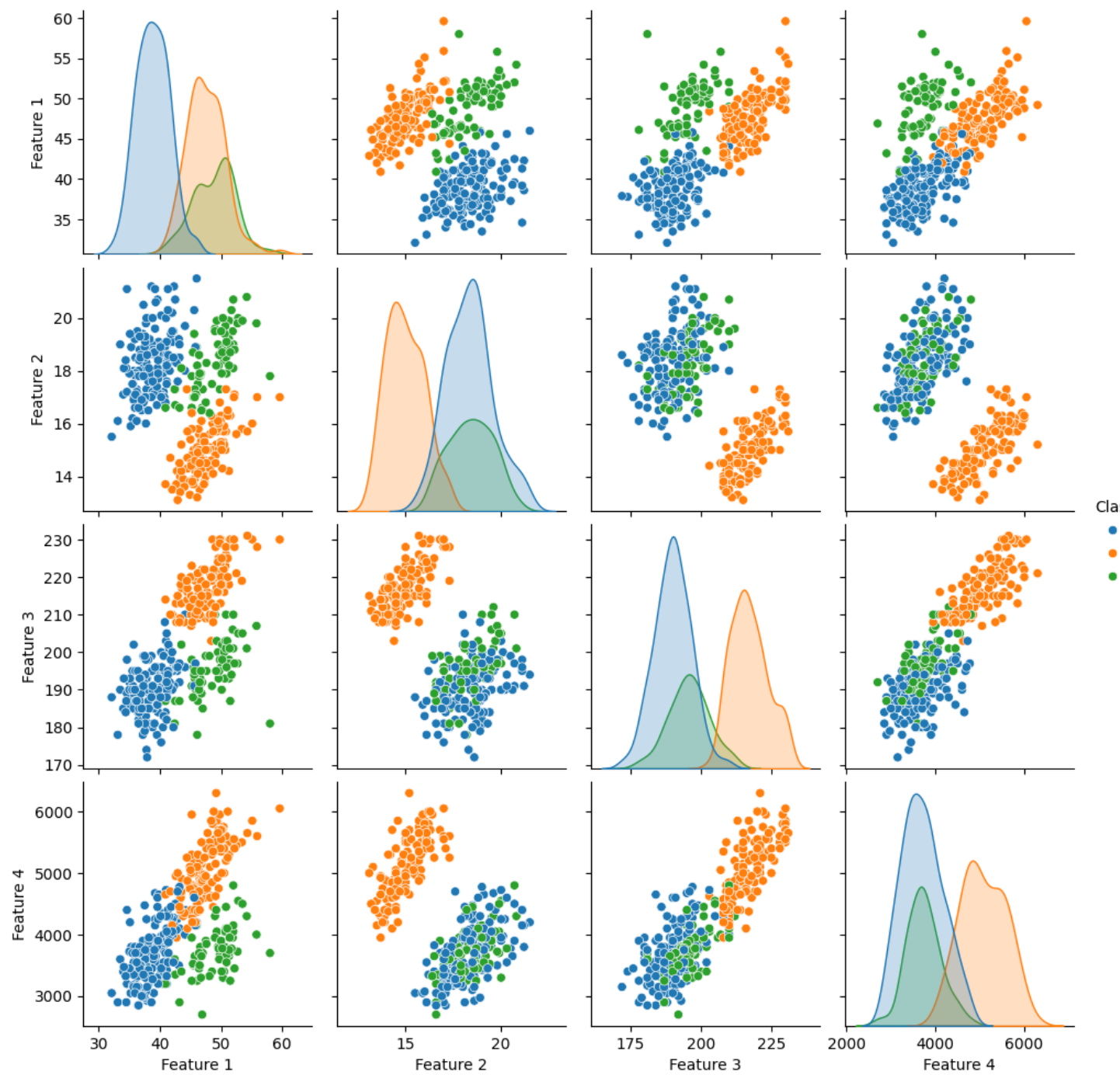
(h) Hexbin plot



(i) Correlation matrix



(j) Pair plot for all features



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